



Adyapan

School

Computer Networks



Duration - 2 months

**Industry
Certification**



Skill India Certified

250+
Partner Companies

From protocols to performance Own networking.

From protocols to connected systems - become industry-ready.

This immersive Computer Networks program takes you beyond foundational concepts into real-world network design and management. Learn network architectures, routing and switching, network security and protocol analysis through hands-on labs and guided simulations. With a strong focus on practical implementation and industry practices, you'll graduate with the ability to design, configure and troubleshoot networks while ensuring secure and efficient communication across systems with confidence.

8

MODULES

30+

PROGRAM OFFERINGS

20,000+

STUDENTS

250+ PARTNERED COMPANIES

ABOUT ADYAPAN SCHOOLS

Where education meets real-world impact

Not just a course - a platform to launch
your career.

Adyapan Schools was built with a single conviction:
learning works best when it happens in the real world.
We partner with top companies, mentors, and industry
platforms to ensure every student graduates with a
portfolio of work that speaks louder than a certificate.

Our programs combine rigorous coursework with live
client projects, giving you the skills and proof-of-work
that employers actually want.

MISSION

To equip ambitious learners with
practitioner-level digital
marketing skills through mentor-
led, project-based education that
bridges the gap between learning
and earning.



VISION

To be India's most trusted
launchpad for the next generation
of marketing leaders – defined
not by degrees but by the real
work.



Everything you need to grow fast

PROGRAM HIGHLIGHTS



Live Industry Projects

Work on campaigns for real brands alongside your coursework. Build portfolio projects that prove your expertise to employers.



1-on-1 Mentorship

Dedicated mentors from Google, Microsoft, Mastercard and more. Get personalized guidance and industry connections.



AI-Powered Marketing

Learn cutting-edge AI tools alongside evergreen fundamentals. Stay ahead of the curve in a rapidly evolving landscape.



Dual Certification

Earn both a Course Completion and Internship Certificate – accredited by Skill India Digital Hub and NSDC.



Internship Guarantee

Graduate with an internship completion certificate from a live brand project. Concrete, resume-ready proof of work.



Industry Network

Join a network of alumni at Amazon, Google, Adobe, Microsoft. Access exclusive hiring events and referral opportunities.

8 weeks. 8 modules. Infinite impact.

WEEK 1

Introduction to Computer Networks

- Discussion of curriculum
- Understand the history of computer networks and how they evolved from early ARPANET to the modern Internet
- Compare circuit switching and packet switching — how each works, their trade-offs, and why packet switching won
- Explore the TCP/IP protocol stack: the five layers, what each layer does, and how they interact end to end
- Understand encapsulation and decapsulation as data moves down and up the protocol stack



WEEK 2

Application Layer Services

- Understand how HTTP works: request-response model, methods, status codes, persistent vs non-persistent connections
- Explore FTP and email protocols (SMTP, POP3, IMAP) and how they transfer files and messages across networks
- Learn how DNS resolves domain names to IP addresses using a hierarchical, distributed database system
- Examine how application layer protocols rely on the transport layer and what service guarantees they expect



WEEK 3

Transport Layer - TCP Fundamentals

- Understand transport layer responsibilities: process-to-process delivery, multiplexing, and demultiplexing using port numbers
- Study TCP connection establishment and closure using the three-way handshake and four-way termination
- Explore TCP's core features: reliable delivery, ordered data transfer, byte-stream service, and error detection
- Learn how TCP congestion control works: slow start, congestion avoidance, fast retransmit, and fast recovery



8 weeks. 8 modules. Infinite impact.

WEEK 4

Flow Control & Congestion Control

- Understand flow control and why it is needed to prevent a fast sender from overwhelming a slow receiver
- Study the sliding window mechanism and how TCP uses receiver advertised window size to regulate data flow
- Analyze congestion control at the network level and the signals TCP uses to detect and respond to congestion
- Compare TCP Tahoe, Reno, and CUBIC and understand how each handles packet loss differently



WEEK 5

Network Layer - IP Addressing & Routing

- Understand IPv4 addressing: address classes, subnetting, CIDR notation, and how addresses are allocated
- Learn how routers use forwarding tables to make hop-by-hop decisions and move packets toward their destination
- Study intra-domain routing protocols: how RIP uses distance vector and how OSPF uses link state algorithms
- Explore inter-domain routing with BGP: how autonomous systems exchange reachability information across the internet



WEEK 6

IP Services - ARP & SNMP

- Understand ARP and how it maps IP addresses to MAC addresses within a local area network segment
- Learn how ARP caching works, what ARP spoofing is, and why it presents a security risk in local networks
- Explore SNMP and how network administrators use it to monitor, manage, and configure network devices remotely
- Understand the role of ICMP in network diagnostics and how tools like ping and traceroute use it



8 weeks. 8 modules. Infinite impact.

WEEK 7

Data Link Layer & Media Access Control

- Understand data link layer services: framing, flow control, and error control between directly connected nodes
- Study error detection and correction techniques: parity bits, checksums, and CRC used in frame transmission
- Explore channel access protocols: TDMA, FDMA, ALOHA, CSMA/CD, and CSMA/CA and when each is used
- Learn how Ethernet framing works and how MAC addresses identify devices on a local area network



WEEK 8

End-to-End Principle

- Understand the end-to-end argument in network design: why intelligence belongs at the endpoints, not the network core
- Review how all five layers work together in a complete communication scenario from browser request to server response
- Examine real-world trade-offs in network design: reliability vs performance, centralization vs distribution, simplicity vs features



WHO THIS IS FOR

This course is perfect for

Students & Career Switchers

Aspiring Network Engineers &
Infrastructure Professionals

Computer Science, IT &
Electronics Students

Working Professionals in IT
Support & System Administration

Cyber Security & Cloud Enthusiasts
(Networking Foundations)

Individuals Preparing for
Networking Certifications

CERTIFICATIONS



ALUMNI NETWORK

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Adobe

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